Tutorial - 5 (SVM) 24/02/2026

Suspicious words (x_1)	Links (x_2)	Attachment (x_3)	class (y)
6	3	1	+1
1	0	0	-1
4	2	1	+1
0	0	0	-1
5	2	1	+1
(w_1)	(w_2)	(w_3)	(Bias : w_0)

spam = +1

not spam = -1

weighted sum $(z) = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$ $f(x)$

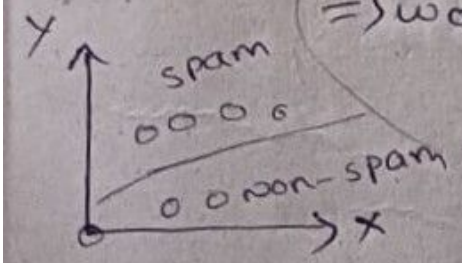
spam :	(4, 5, 6) ✓	(2, 3)	(1)
non-spam :	(0, 1)	(0)	(0)

Initialize : w_2 & $w_3 = 0$

spam / non-spam $= f(x) = 0$ — decision Boundary

If $f(x) = 0 \Rightarrow w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 = 0$

$\Rightarrow w_0 + w_1 x_1 = 0 \therefore (w_2 \,\&\, w_3 = 0)$



$x_1 = 2$; now, $w_1 = 1$

$w_0 + w_1 x_1 = 0$

$w_0 = 0 - w_1 x_1$

$w_0 = 0 - 1(2)$

$\boxed{w_0 = -2}$

Now, weighted sum $f(x) = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 \ldots$

Now calculate (or) check:

$y_i (x_1 w_1 + x_2 w_2 + x_3 w_3 + w_0) \geq 1$ perfect classification

< 1 classification fails.

$f(x_1) = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 - 2$

$= -2 + 1(6) + 0(3) + 0(1) - 2$

$= -2 + 6 + 0 + 0 - 2$

$= -2 + 6 - 2$

$= 6 - 4$

$= 2$

<u>Check $y_i (x_1 - 2)$</u>

$\underline{y_1}$

$\Rightarrow y_1 (x_1 - 2)$

$\Rightarrow 1 (6 - 2)$

$= 4 > 1$. so perfect classification

$\underline{y_2}$

$y_2 (x_2 - 2)$

$-1 (1 - 2)$

$-1(-1)$

$= 1 > 1$ perfect. C

$\underline{y_3}$

$y_3 (x_3 - 2)$

$1 (4 - 2)$

$2 > 1$ P. C

$\underline{y_4}$

$y_4 (x_4 - 2)$

$-1 (0 - 2)$

$= 2 > 1$ P. C

$\underline{y_5}$

$y_5 (x_5 - 2)$

$1 (5 - 2)$

$3 > 1$ P. C

$\hat{y}$

≥ 1 (P.F)

≥ 1 (P.F)

≥ 1 (P.F)

≥ 1 (P.F)

≥ 1 (P.F)

Cost or Loss function:

$$L = \frac{1}{2}\|w\|^2 - \sum_{i=1}^{N} c_i\left[y_i(wx + \overset{w_0}{b}) - 1\right]$$

$$= \frac{1}{2}(w_1^2 + w_2^2 + w_3^2) - \sum_{i=1}^{N} c_i \text{ error.}$$

=) Updated weights

If P.C => $y_i(w_1x_1 + w_2x_2 + \dots) \geq 1$

$C \to 0 \Rightarrow L = \frac{1}{2}w^2$

$$w^{new} = w^{old} - \eta\frac{\partial L}{\partial w}$$

$$= w - \eta w$$

$$w_1^{new} = w_1^{old} - \eta w_1^{old}$$

$$= 1 - 1(1)$$

$$= 1 - 1$$

$$= 0$$

$\eta = 0.1$

$$= 1 - 1(0.1)$$

$$= 1 - 0.1$$

$$= 0.9$$

$$w_0^{new} = w_0^{old} - \eta w_0^{old}$$

$$= -2 - 1(-2)$$

$$= -2 + 2$$

$$= 0$$

$\eta = 0.1$

$$= -2 - 0.1(-2)$$

$$= -2 + 0.2$$

$$= -1.8$$

$$\text{Margine} = \frac{2}{\|w\|}$$

$$= \frac{2}{\sqrt{w_1^2 + w_2^2 + w_3^2 + \dots}}$$

$$w^{new} = w - \eta \frac{\partial L}{\partial w}$$ If not P.C $\Rightarrow$ $y_i(w_1 x_1 + w_2 x_2 + \cdots) < 1$

$$= w - \eta \left[w - \sum_i C_i y_i x_i \right]$$

$$= (1-\eta) w + \eta \sum_i C_i y_i x_i$$

$$w_1 = \sum_i C_i y_i x_i = C \sum_i y_i x_i$$

$C = 1 \Rightarrow$ $y_1 x_{11} + y_2 x_{12} + y_3 x_{13} + y_4 x_{14} + y_5 x_{15}$

Good Classification $\rightarrow$ Bias not chained

Bad Classification $\rightarrow$ W & B

$w_1 = (1(6) + (-1)(1) + (+1)(4) + (-1)(0) + (+1)(5))$

$w_1^{new} = 6 - 1 + 4 + 0 + 5 = 5 + 9 = 14.$

$w_2 = (y_1 x_{21} + y_2 x_{22} + y_3 x_{23} + y_4 x_{24} + y_5 x_{25})$

$w_2^{new} = (1(3) + (-1)(0) + (+1)(2) + (-1)(0) + (+1)(2))$

$= 3 + 0 + 2 + 0 + 2 = 7$

$w_3^{new} = (1(1) + (-1)(0) + (+1)(2) + (-1)(0) + (+1)(1))$

$= 1 + 0 + 1 + 0 + 1 = 3$